2.Actividad Clustering Gerarquico

November 4, 2026

[1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import os
```

[2]:
```
#Importamos la base
data = pd.read_excel("Amazon.xlsx")
data.head()
```

[2]:

	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	Servicio Retorno	Tamano Paquete	Calidad Producto	Numero Estrellas
0	Adam	205	3	345	235	24	23	26	21	17
1	Anna	9	15	315	33	25	4	42	215	28
2	Bernard	17	26	285	3	43	27	41	26	33
3	Edward	135	5	355	295	18	23	39	195	17
4	Emilia	3	45	48	39	34	46	225	34	43

[3]:
```
#Normalizamos los datos
from sklearn.preprocessing import normalize

data_numerica = data.select_dtypes(include='number')
data_scaled = normalize(data_numerica)
data_scaled
```

```
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```

[4]:
```
#Convertir los datos normalizados en data Frame
data_scaled = pd.DataFrame(data_scaled, columns=data_numerica.columns)
data_scaled.head()
```

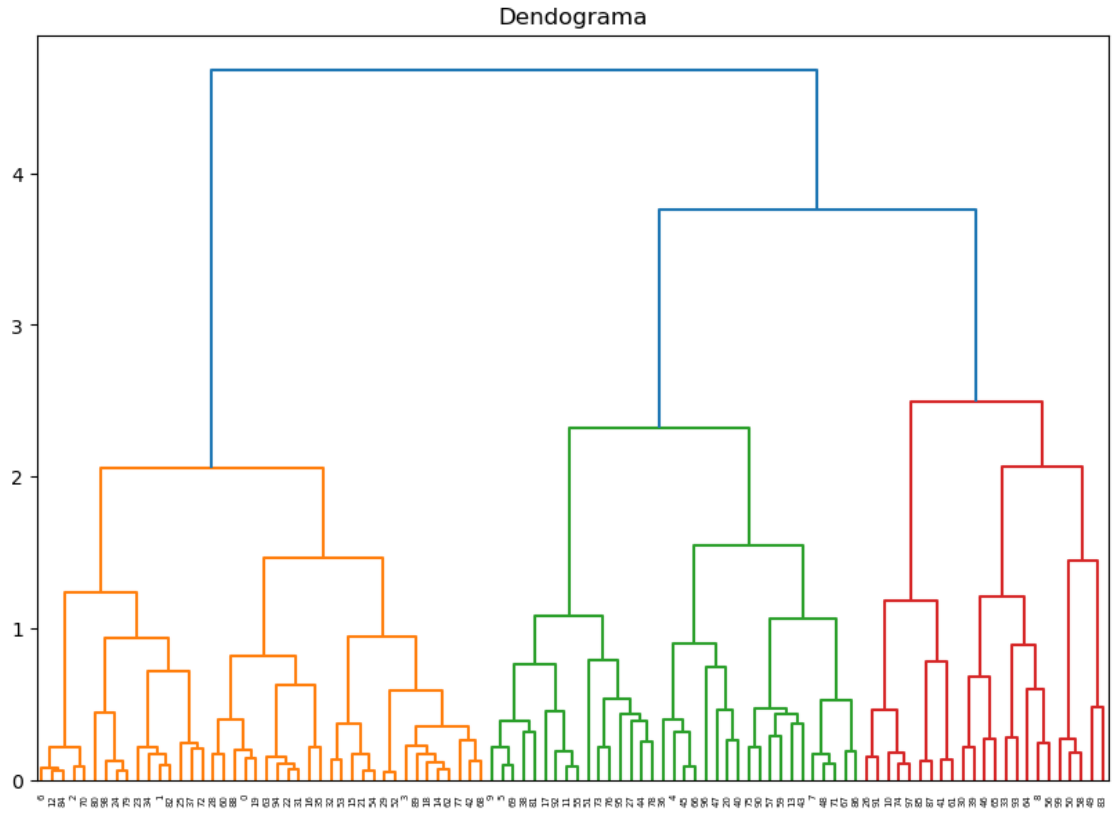
[4]:

	Velocidad Entrega	Precio	Durabilidad	Imagen Producto	Valor Educativo	\
0	0.438263	0.006414	0.737565	0.502399	0.051309	
1	0.023235	0.038725	0.813234	0.085196	0.064542	
2	0.057235	0.087535	0.959520	0.010100	0.144770	
3	0.258856	0.009587	0.680696	0.565649	0.034514	
4	0.011975	0.179625	0.191600	0.155675	0.135717	

	Servicio Retorno	Tamano Paquete	Calidad Producto	Numero Estrellas
0	0.049171	0.055585	0.044895	0.036344
1	0.010327	0.108431	0.555065	0.072287
2	0.090902	0.138036	0.087535	0.111102
3	0.044101	0.074781	0.373904	0.032597
4	0.183617	0.898127	0.135717	0.171642

[5]:
```
#Construir el Dendograma
import scipy.cluster.hierarchy as shc

plt.figure(figsize=(10,7))
plt.title("Dendograma")
dend = shc.dendrogram(shc.linkage(data_scaled, method='ward'))
```



[6]:
```
#Calcular el número de clústeres del ejemplo
colores_unicos = set(dend['color_list'])
num_clusters_optimo = len(colores_unicos)-1
num_clusters_optimo
```

[6]: 3

[7]:
```
# Análisis gráfico con PCA
campos = data_scaled
from sklearn import decomposition
pca = decomposition.PCA(n_components = 2)
pca.fit(campos)
campos = pca.transform(campos)
campos
```

[7]:
```
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[-0.42217536, -0.03448406],
[-0.16141571,  0.48420739],
[-0.65108758,  0.4197626 ],
[-0.19657428, -0.70259997],
[-0.66940341,  0.17418122],
[-0.50314121, -0.04478847],
[-0.37827002, -0.22768866],
[-0.09257978, -0.29000069],
[-0.15947025,  0.08591885],
[ 0.16414836, -0.33801587],
```

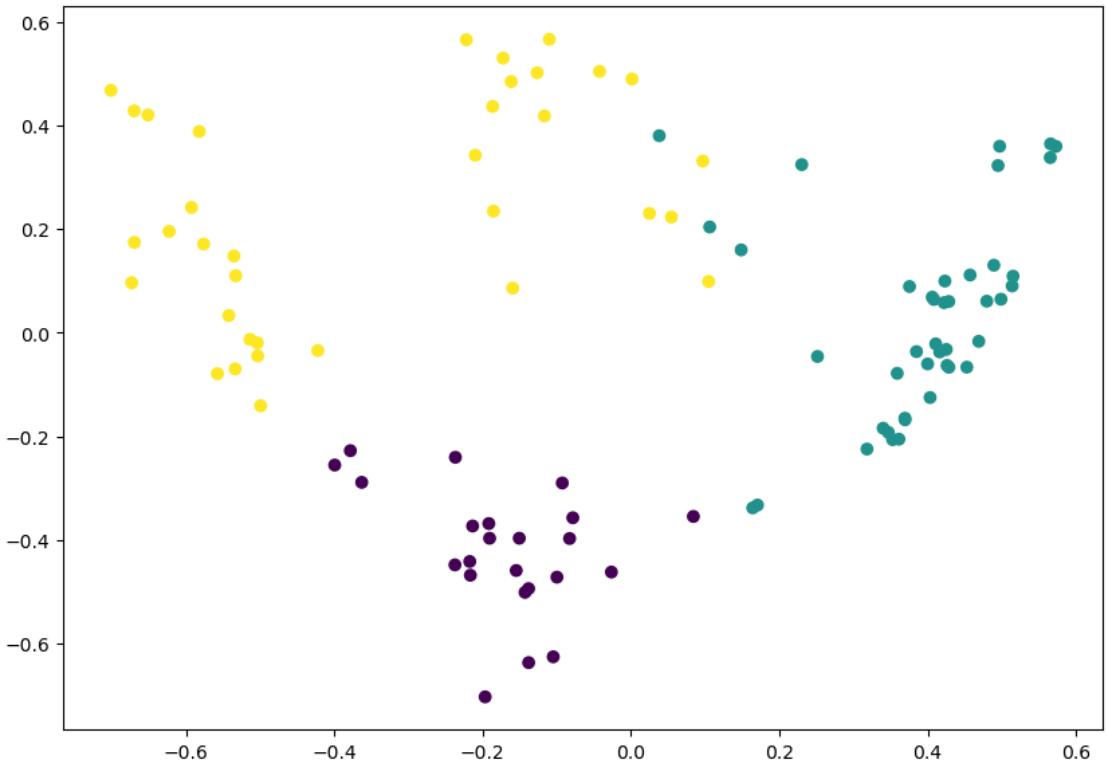
```
[ 0.35892963, -0.07853561],
[ 0.49880502,  0.06450716],
[-0.04236318,  0.50393996],
[-0.13801031, -0.49375593],
[-0.57613865,  0.17095284],
[-0.0783886 , -0.35710609],
[-0.67312488,  0.09618175],
[ 0.47969491,  0.06068297],
[-0.39939783, -0.25539679],
[ 0.3695398 , -0.16814344],
[ 0.42508059, -0.03242069],
[-0.02636687, -0.46187498],
[-0.10479979, -0.62527833],
[-0.66976445,  0.42777742],
[-0.55740005, -0.0791324 ],
[ 0.36929226, -0.16505085],
[-0.12646304,  0.50126007],
[ 0.49481837,  0.32229269],
[-0.50365854, -0.01974903],
[ 0.23024785,  0.32414928],
[ 0.02497426,  0.23000985],
[-0.15056994, -0.39659029],
[-0.59247292,  0.24149892],
[ 0.10475686,  0.09868653],
[ 0.45279472, -0.0665795 ],
[-0.2099264 ,  0.34219382],
[ 0.41622673, -0.03730307],
[ 0.14855054,  0.1595983 ],
[-0.11014101,  0.56579396],
[ 0.40616865,  0.06841998],
[-0.23672835, -0.24043862],
[ 0.56568028,  0.36408458],
[-0.23730948, -0.44802811],
[-0.49928987, -0.14100592],
[-0.21649152, -0.46796438],
[ 0.42855949, -0.06683056],
[ 0.31814994, -0.22452555],
[-0.53277044,  0.10971321],
[-0.15466959, -0.45884497],
[-0.11664927,  0.41801606],
[-0.21734196, -0.44145587],
[ 0.45720098,  0.11118217],
[-0.18527671,  0.23439125],
[-0.53527538,  0.14792444],
[-0.19164058, -0.36824856],
[ 0.38487326, -0.03658414],
[ 0.08413049, -0.3546116 ]])
```

[8]:
```
#Aplicar clustering jerárquico
from sklearn.cluster import AgglomerativeClustering
cluster = AgglomerativeClustering(n_clusters=3, linkage='ward')
grupos = cluster.fit_predict(data_scaled)
grupos
```

[8]:
```
array([1, 1, 1, 1, 2, 2, 1, 2, 0, 2, 0, 2, 1, 2, 1, 1, 1, 2, 1, 1, 2, 1,
       1, 1, 1, 1, 0, 2, 1, 1, 0, 1, 1, 0, 1, 1, 2, 1, 2, 0, 2, 0, 1, 2,
       2, 2, 0, 2, 2, 0, 0, 2, 1, 1, 1, 2, 0, 2, 0, 2, 1, 0, 1, 1, 0, 0,
       2, 2, 1, 2, 1, 2, 1, 2, 0, 2, 2, 1, 2, 1, 1, 2, 1, 0, 1, 0, 2, 0,
       1, 1, 2, 0, 2, 0, 1, 2, 2, 0, 1, 0])
```

[9]:
```
#Graficar los grupos con PCA
plt.figure(figsize=(10, 7))
plt.scatter(campos[:,0], campos[:,1], c=cluster.labels_)
```

[9]:
```
<matplotlib.collections.PathCollection at 0x7043e715f830>
```



[10]:
```
#Crear el DataFrame con los grupos
dataframe = pd.DataFrame(grupos, columns=['grupo'])
dataframe
```

[10]:
```
    grupo
0       1
1       1
2       1
3       1
4       2
..    …
95      2
96      2
97      0
98      1
99      0

[100 rows x 1 columns]
```

[11]:
```
#Unir los grupos con la base original
dataframe2 = pd.concat([data.reset_index(drop=True), dataframe], axis=1,␣
 ↪join='inner')
dataframe2
```

[11]:
```
   Unnamed: 0  Velocidad Entrega  Precio  Durabilidad  Imagen Producto  \
0        Adam                205       3          345              235
1        Anna                  9      15          315               33
2     Bernard                 17      26          285                3
3      Edward                135       5          355              295
4      Emilia                  3      45           48               39
..        …                  …     …          …              …
95     Teofan                  3       8           32               25
96     Teofil                305      25           46               24
97    Teofila                  1      14           26               25
98       Teon                155      11          335               34
99     Teresa                125       9           45               25

    Valor Educativo  Servicio Retorno  Tamano Paquete  Calidad Producto  \
0                24                23              26                21
1                25                 4              42               215
2                43                27              41                26
3                18                23              39               195
4                34                46             225                34
..              …                 …             …              …
95                7                21              42                17
96               33                28             355                26
97               24                27              42               185
98               26                29              42               215
99               22                 3               3                22

    Numero Estrellas  grupo
```

```
0                  17      1
1                  28      1
2                  33      1
3                  17      1
4                  43      2
..                 …     …
95                  1      2
96                 45      2
97                 23      0
98                 27      1
99                 18      0

[100 rows x 11 columns]
```

[12]:
```
data.columns
```

[12]:
```
Index(['Unnamed: 0', 'Velocidad Entrega', 'Precio', 'Durabilidad',
       'Imagen Producto', 'Valor Educativo', 'Servicio Retorno',
       'Tamano Paquete', 'Calidad Producto', 'Numero Estrellas'],
      dtype='object')
```

[13]:
```
#Identificar los clientes de cada clúster
dataframe2.index = dataframe2['Unnamed: 0']
dataframe2
```

[13]:
```
           Unnamed: 0  Velocidad Entrega  Precio  Durabilidad  \
Unnamed: 0
Adam             Adam                205       3          345
Anna             Anna                  9      15          315
Bernard       Bernard                 17      26          285
Edward         Edward                135       5          355
Emilia         Emilia                  3      45           48
…                 …                  …      …          …
Teofan         Teofan                  3       8           32
Teofil         Teofil                305      25           46
Teofila       Teofila                  1      14           26
Teon             Teon                155      11          335
Teresa         Teresa                125       9           45

             Imagen Producto  Valor Educativo  Servicio Retorno  \
Unnamed: 0
Adam                     235               24                23
Anna                      33               25                 4
Bernard                    3               43                27
Edward                   295               18                23
Emilia                    39               34                46
…                        …                 …                 …
```

```
Teofan                    25                 7                 21
Teofil                    24                33                 28
Teofila                   25                24                 27
Teon                      34                26                 29
Teresa                    25                22                  3

            Tamano Paquete  Calidad Producto  Numero Estrellas  grupo
Unnamed: 0
Adam                    26                21                17      1
Anna                    42               215                28      1
Bernard                 41                26                33      1
Edward                  39               195                17      1
Emilia                 225                34                43      2
…                      …                 …                 …     …
Teofan                  42                17                 1      2
Teofil                 355                26                45      2
Teofila                 42               185                23      0
Teon                    42               215                27      1
Teresa                   3                22                18      0

[100 rows x 11 columns]
```

[15]:
```
#identificar a Salome
dataframe2.loc['Salome']
```

[15]:
```
Unnamed: 0           Salome
Velocidad Entrega        17
Precio                   23
Durabilidad             275
Imagen Producto          41
Valor Educativo           4
Servicio Retorno         44
Tamano Paquete          315
Calidad Producto         28
Numero Estrellas         32
grupo                     2
Name: Salome, dtype: object
```

[17]:
```
#Clientes de su mismo grupo
grupo_salome = dataframe2.loc['Salome', 'grupo']
dataframe2[dataframe2['grupo'] == grupo_salome]
```

[17]:
```
           Unnamed: 0  Velocidad Entrega  Precio  Durabilidad  \
Unnamed: 0
Emilia         Emilia                  3      45           48
Fabian         Fabian                 95     165          395
Frank           Frank                 65      21           31
```

Gabriel	Gabriel	2	175	325
Henry	Henry	195	11	455
Isabelle	Isabelle	185	75	43
Eugenia	Eugenia	245	9	385
Evdokia	Evdokia	165	45	43
Florence	Florence	26	65	485
Jeremiah	Jeremiah	18	2	29
Joachim	Joachim	0	105	345
Santiago	Santiago	95	17	38
Justin	Justin	25	65	43
Kalyna	Kalyna	1	13	325
Larissa	Larissa	25	125	47
Leon	Leon	17	195	28
Leonard	Leonard	29	1	44
Leo	Leo	13	24	41
Magdalyna	Magdalyna	145	13	385
Marcel	Marcel	27	125	48
Maria	Maria	115	225	4
Maryna	Maryna	21	125	46
Matthew	Matthew	8	225	32
Maya	Maya	115	185	415
Melania	Melania	28	11	41
Michael	Michael	26	65	455
Mina	Mina	21	12	47
Monica	Monica	19	4	415
Myron	Myron	5	95	355
Salome	Salome	17	23	275
Sebastian	Sebastian	12	145	42
Susanna	Susanna	15	14	39
Sylvester	Sylvester	155	21	255
Teofan	Teofan	3	8	32
Teofil	Teofil	305	25	46

Unnamed: 0	Imagen Producto	Valor Educativo	Servicio Retorno	\
Emilia	39	34	46	
Fabian	24	26	19	
Frank	255	28	22	
Gabriel	3	37	32	
Henry	23	3	25	
Isabelle	285	27	37	
Eugenia	215	34	15	
Evdokia	2	21	18	
Florence	305	32	39	
Jeremiah	29	37	25	
Joachim	27	11	26	
Santiago	23	26	25	

Justin	235	31	25
Kalyna	185	24	17
Larissa	23	37	14
Leon	28	36	23
Leonard	225	3	24
Leo	25	36	25
Magdalyna	35	28	36
Marcel	275	4	3
Maria	235	33	22
Maryna	31	33	39
Matthew	265	3	25
Maya	26	3	23
Melania	155	4	16
Michael	225	33	27
Mina	245	32	27
Monica	305	22	26
Myron	225	15	31
Salome	41	4	44
Sebastian	295	27	27
Susanna	355	3	38
Sylvester	39	36	4
Teofan	25	7	21
Teofil	24	33	28

Unnamed: 0	Tamano Paquete	Calidad Producto	Numero Estrellas	grupo
Emilia	225	34	43	2
Fabian	485	22	3	2
Frank	345	215	29	2
Gabriel	435	27	39	2
Henry	415	25	32	2
Isabelle	335	25	23	2
Eugenia	295	28	25	2
Evdokia	315	225	26	2
Florence	335	29	39	2
Jeremiah	465	24	29	2
Joachim	445	195	14	2
Santiago	385	185	25	2
Justin	185	24	33	2
Kalyna	425	16	23	2
Larissa	315	3	39	2
Leon	455	235	28	2
Leonard	335	245	42	2
Leo	45	26	38	2
Magdalyna	385	21	32	2
Marcel	385	3	5	2
Maria	435	25	35	2

```
Maryna                365                3                44       2
Matthew               355              225                31       2
Maya                  455               24                34       2
Melania               265              195                 4       2
Michael               365              255                45       2
Mina                  425               26                34       2
Monica                265              255                27       2
Myron                 495              165                24       2
Salome                315               28                32       2
Sebastian             335              275                36       2
Susanna               395               22                34       2
Sylvester             295               26                28       2
Teofan                 42               17                 1       2
Teofil                355               26                45       2
```

[19]:
```
#identificar cluster steph
dataframe2.loc['Stephania']
```

[19]:
```
Unnamed: 0           Stephania
Velocidad Entrega          215
Precio                     125
Durabilidad                465
Imagen Producto            315
Valor Educativo             34
Servicio Retorno             4
Tamano Paquete              37
Calidad Producto           305
Numero Estrellas            45
grupo                        1
Name: Stephania, dtype: object
```

[20]:
```
grupo_stephania = dataframe2.loc['Stephania', 'grupo']
dataframe2[dataframe2['grupo'] == grupo_stephania]
```

[20]:
```
              Unnamed: 0  Velocidad Entrega  Precio  Durabilidad  \
Unnamed: 0
Adam                Adam                205       3          345
Anna                Anna                  9      15          315
Bernard          Bernard                 17      26          285
Edward            Edward                135       5          355
Philip            Philip                 23      12          475
Irene              Irene                 14       7          405
Isidore          Isidore                235      65          495
Joseph            Joseph                 17       1          485
Eugene            Eugene                 16     205          285
Eunice            Eunice                265       7          485
Eva                  Eva                235      65          495
```

Fedir	Fedir	17	2	415
Felix	Felix	15	2	455
Fialka	Fialka	12	75	335
Flavia	Flavia	255	7	435
Flora	Flora	23	105	395
Hannah	Hannah	175	14	495
Helen	Helen	205	185	295
Hilary	Hilary	14	19	445
Lourdes	Lourdes	26	1	465
Ivan	Ivan	12	5	385
Jacob	Jacob	9	165	375
Jervis	Jervis	2	45	455
Judith	Judith	245	115	465
Louise	Louise	225	205	315
Lubomyr	Lubomyr	14	12	335
Lydia	Lydia	19	4	435
Marian	Marian	155	95	495
Markian	Markian	205	55	465
Marko	Marko	15	19	275
Maura	Maura	265	85	425
Maximillian	Maximillian	18	27	295
Methodius	Methodius	18	11	495
Mykyta	Mykyta	165	13	485
Myroslav	Myroslav	225	8	435
Myroslava	Myroslava	275	9	435
Samuel	Samuel	8	14	305
Sarah	Sarah	13	15	425
Stephan	Stephan	145	6	365
Stephania	Stephania	215	125	465
Theodore	Theodore	2	25	335
Teon	Teon	155	11	335

Unnamed: 0	Imagen Producto	Valor Educativo	Servicio Retorno \
Adam	235	24	23
Anna	33	25	4
Bernard	3	43	27
Edward	295	18	23
Philip	33	35	45
Irene	19	21	14
Isidore	335	3	26
Joseph	235	27	17
Eugene	255	36	29
Eunice	305	33	39
Eva	335	3	26
Fedir	125	12	17
Felix	355	35	34

Fialka	24	19	25
Flavia	24	33	26
Flora	29	34	28
Hannah	175	31	17
Helen	275	39	3
Hilary	345	33	32
Lourdes	295	37	24
Ivan	17	17	11
Jacob	225	25	24
Jervis	27	24	26
Judith	225	36	13
Louise	295	43	34
Lubomyr	245	25	26
Lydia	145	16	21
Marian	225	26	31
Markian	275	25	27
Marko	245	34	26
Maura	185	35	19
Maximillian	31	45	29
Methodius	24	29	19
Mykyta	165	29	15
Myroslav	23	31	21
Myroslava	19	36	21
Samuel	32	23	38
Sarah	3	28	28
Stephan	305	2	25
Stephania	315	34	4
Theodore	225	22	21
Teon	34	26	29

Unnamed: 0	Tamano Paquete	Calidad Producto	Numero Estrellas	grupo
Adam	26	21	17	1
Anna	42	215	28	1
Bernard	41	26	33	1
Edward	39	195	17	1
Philip	38	29	31	1
Irene	33	22	24	1
Isidore	34	295	39	1
Joseph	24	235	34	1
Eugene	31	22	23	1
Eunice	34	295	39	1
Eva	34	3	4	1
Fedir	26	165	2	1
Felix	42	26	4	1
Fialka	36	185	21	1
Flavia	19	245	34	1

```
Flora                235              295               34      1
Hannah                27               27               34      1
Helen                 42              255               31      1
Hilary                41               25               38      1
Lourdes               23              305               45      1
Ivan                  31              205                2      1
Jacob                 38               18               24      1
Jervis               365              255               31      1
Judith                31              295               38      1
Louise                44              275               35      1
Lubomyr               46              185               17      1
Lydia                 28              185               24      1
Marian                19               24               39      1
Markian               37              265               32      1
Marko                  3               21               21      1
Maura                 24              215               43      1
Maximillian           42               27               35      1
Methodius            245              245               36      1
Mykyta                26              255               32      1
Myroslav              34              255               41      1
Myroslava            245              225               44      1
Samuel                41              205               26      1
Sarah                 34               28               38      1
Stephan                4                2               19      1
Stephania             37              305               45      1
Theodore              25                2               16      1
Teon                  42              215               27      1
```

[22]:
```
dataframe2.loc['Lydia']
```

[22]:
```
Unnamed: 0           Lydia
Velocidad Entrega       19
Precio                   4
Durabilidad            435
Imagen Producto        145
Valor Educativo         16
Servicio Retorno        21
Tamano Paquete          28
Calidad Producto       185
Numero Estrellas        24
grupo                    1
Name: Lydia, dtype: object
```

[23]:
```
grupo_lydia = dataframe2.loc['Lydia', 'grupo']
dataframe2[dataframe2['grupo'] == grupo_lydia]
```

[23]:

Unnamed: 0	Unnamed: 0	Velocidad Entrega	Precio	Durabilidad	\
Adam	Adam	205	3	345	
Anna	Anna	9	15	315	
Bernard	Bernard	17	26	285	
Edward	Edward	135	5	355	
Philip	Philip	23	12	475	
Irene	Irene	14	7	405	
Isidore	Isidore	235	65	495	
Joseph	Joseph	17	1	485	
Eugene	Eugene	16	205	285	
Eunice	Eunice	265	7	485	
Eva	Eva	235	65	495	
Fedir	Fedir	17	2	415	
Felix	Felix	15	2	455	
Fialka	Fialka	12	75	335	
Flavia	Flavia	255	7	435	
Flora	Flora	23	105	395	
Hannah	Hannah	175	14	495	
Helen	Helen	205	185	295	
Hilary	Hilary	14	19	445	
Lourdes	Lourdes	26	1	465	
Ivan	Ivan	12	5	385	
Jacob	Jacob	9	165	375	
Jervis	Jervis	2	45	455	
Judith	Judith	245	115	465	
Louise	Louise	225	205	315	
Lubomyr	Lubomyr	14	12	335	
Lydia	Lydia	19	4	435	
Marian	Marian	155	95	495	
Markian	Markian	205	55	465	
Marko	Marko	15	19	275	
Maura	Maura	265	85	425	
Maximillian	Maximillian	18	27	295	
Methodius	Methodius	18	11	495	
Mykyta	Mykyta	165	13	485	
Myroslav	Myroslav	225	8	435	
Myroslava	Myroslava	275	9	435	
Samuel	Samuel	8	14	305	
Sarah	Sarah	13	15	425	
Stephan	Stephan	145	6	365	
Stephania	Stephania	215	125	465	
Theodore	Theodore	2	25	335	
Teon	Teon	155	11	335	

	Imagen Producto	Valor Educativo	Servicio Retorno	\
Unnamed: 0				

Adam	235	24	23
Anna	33	25	4
Bernard	3	43	27
Edward	295	18	23
Philip	33	35	45
Irene	19	21	14
Isidore	335	3	26
Joseph	235	27	17
Eugene	255	36	29
Eunice	305	33	39
Eva	335	3	26
Fedir	125	12	17
Felix	355	35	34
Fialka	24	19	25
Flavia	24	33	26
Flora	29	34	28
Hannah	175	31	17
Helen	275	39	3
Hilary	345	33	32
Lourdes	295	37	24
Ivan	17	17	11
Jacob	225	25	24
Jervis	27	24	26
Judith	225	36	13
Louise	295	43	34
Lubomyr	245	25	26
Lydia	145	16	21
Marian	225	26	31
Markian	275	25	27
Marko	245	34	26
Maura	185	35	19
Maximillian	31	45	29
Methodius	24	29	19
Mykyta	165	29	15
Myroslav	23	31	21
Myroslava	19	36	21
Samuel	32	23	38
Sarah	3	28	28
Stephan	305	2	25
Stephania	315	34	4
Theodore	225	22	21
Teon	34	26	29

Unnamed: 0	Tamano Paquete	Calidad Producto	Numero Estrellas	grupo
Adam	26	21	17	1
Anna	42	215	28	1

Bernard	41	26	33	1
Edward	39	195	17	1
Philip	38	29	31	1
Irene	33	22	24	1
Isidore	34	295	39	1
Joseph	24	235	34	1
Eugene	31	22	23	1
Eunice	34	295	39	1
Eva	34	3	4	1
Fedir	26	165	2	1
Felix	42	26	4	1
Fialka	36	185	21	1
Flavia	19	245	34	1
Flora	235	295	34	1
Hannah	27	27	34	1
Helen	42	255	31	1
Hilary	41	25	38	1
Lourdes	23	305	45	1
Ivan	31	205	2	1
Jacob	38	18	24	1
Jervis	365	255	31	1
Judith	31	295	38	1
Louise	44	275	35	1
Lubomyr	46	185	17	1
Lydia	28	185	24	1
Marian	19	24	39	1
Markian	37	265	32	1
Marko	3	21	21	1
Maura	24	215	43	1
Maximillian	42	27	35	1
Methodius	245	245	36	1
Mykyta	26	255	32	1
Myroslav	34	255	41	1
Myroslava	245	225	44	1
Samuel	41	205	26	1
Sarah	34	28	38	1
Stephan	4	2	19	1
Stephania	37	305	45	1
Theodore	25	2	16	1
Teon	42	215	27	1

[24]:
```
print("Clientes similares a Salomé:")
print(dataframe2[dataframe2['grupo'] == grupo_salome].index.tolist())

print("Clientes similares a Stephanía:")
print(dataframe2[dataframe2['grupo'] == grupo_stephania].index.tolist())
```

```
print("Clientes similares a Lydia:")
print(dataframe2[dataframe2['grupo'] == grupo_lydia].index.tolist())
```

```
Clientes similares a Salomé:
['Emilia', 'Fabian', 'Frank', 'Gabriel', 'Henry', 'Isabelle', 'Eugenia',
'Evdokia', 'Florence', 'Jeremiah', 'Joachim', 'Santiago', 'Justin', 'Kalyna',
'Larissa', 'Leon', 'Leonard', 'Leo', 'Magdalyna', 'Marcel', 'Maria', 'Maryna',
'Matthew', 'Maya', 'Melania', 'Michael', 'Mina', 'Monica', 'Myron', 'Salome',
'Sebastian', 'Susanna', 'Sylvester', 'Teofan', 'Teofil']
Clientes similares a Stephanía:
['Adam', 'Anna', 'Bernard', 'Edward', 'Philip', 'Irene', 'Isidore', 'Joseph',
'Eugene', 'Eunice', 'Eva', 'Fedir', 'Felix', 'Fialka', 'Flavia', 'Flora',
'Hannah', 'Helen', 'Hilary', 'Lourdes', 'Ivan', 'Jacob', 'Jervis', 'Judith',
'Louise', 'Lubomyr', 'Lydia', 'Marian', 'Markian', 'Marko', 'Maura',
'Maximillian', 'Methodius', 'Mykyta', 'Myroslav', 'Myroslava', 'Samuel',
'Sarah', 'Stephan', 'Stephania', 'Theodore', 'Teon']
Clientes similares a Lydia:
['Adam', 'Anna', 'Bernard', 'Edward', 'Philip', 'Irene', 'Isidore', 'Joseph',
'Eugene', 'Eunice', 'Eva', 'Fedir', 'Felix', 'Fialka', 'Flavia', 'Flora',
'Hannah', 'Helen', 'Hilary', 'Lourdes', 'Ivan', 'Jacob', 'Jervis', 'Judith',
'Louise', 'Lubomyr', 'Lydia', 'Marian', 'Markian', 'Marko', 'Maura',
'Maximillian', 'Methodius', 'Mykyta', 'Myroslav', 'Myroslava', 'Samuel',
'Sarah', 'Stephan', 'Stephania', 'Theodore', 'Teon']
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